

Exercise 01: Dimensionality Reduction with PCA and PCA+SVD

Given Dataset: Consider a 2D dataset with $N = 4$ samples represented by matrix $X \in \mathbb{R}^{2 \times 4}$:

$$X = \begin{bmatrix} 2 & 3 & 5 & 6 \\ 1 & 5 & 4 & 6 \end{bmatrix}$$

- **Q1: Standard PCA via Covariance Matrix**

Implement the standard PCA algorithm step-by-step (8 steps). For each step, print the computed and visualize numerical results.

- **Q2: PCA using SVD**

Implement PCA via direct Singular Value Decomposition (SVD) step-by-step (8 steps). For each step, print the computed and visualize numerical results.

- **Q3: Visualization**

Plot original points X , center μ , and reconstructed points $X_{\text{reconstructed}}$ with the red dashed PC1 axis and dotted gray orthogonal projection lines connecting each X_i to its projection.

(Subplot 1: PCA; Subplot 2: PCA+SVD)

Exercise 02: Linear Discriminant Analysis (LDA)

Given Dataset: Sample 2 classes (setosa and versicolor) from the Iris dataset, selecting 5 random samples per class ($N = 10$, $d = 4$).

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	4.3000	3.0000	1.1000	0.1000	setosa
1	5.1000	3.4000	1.5000	0.2000	setosa
2	4.8000	3.1000	1.6000	0.2000	setosa
3	4.8000	3.0000	1.4000	0.3000	setosa
4	5.1000	3.5000	1.4000	0.3000	setosa
5	6.1000	2.9000	4.7000	1.4000	versicolor
6	5.5000	2.5000	4.0000	1.3000	versicolor
7	5.5000	2.4000	3.8000	1.1000	versicolor
8	5.7000	3.0000	4.2000	1.2000	versicolor
9	5.8000	2.7000	4.1000	1.0000	versicolor

Requirement: Implement the Linear Discriminant Analysis (LDA) algorithm step-by-step.

- Step 1: Calculate each class mean vector μ_i , the within-class scatter matrices S_i , and the overall within-class scatter matrix S_W . Print the values of μ_i , S_i , and S_W .
- Step 2: Compute the global mean vector μ and the between-class scatter matrix S_B . Print the values of μ and S_B .
- Step 3: Solve the generalized eigenvalue problem for $S_W^{-1}S_B$ to find eigenvalues (λ_i) and eigenvectors (w_i). Print the values of λ_i and w_i .
- Step 4: Select the leading eigenvector to construct the projection matrix W and compute the 1D transformed coordinates ($y = W^T x$). Print the values of W and the transformed coordinates y .
- Step 5: Plot the 1D projection results.